

Classifying Suicide-Related Communication on Reddit

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Subtle linguistic signs of suicidal ideation are present in postings on Reddit. It is impossible to check all the content by moderators that would result in high-risk posts being overlooked. The study examines language bias on NLP suicide-risk classification based on posts that have been labeled as *Supportive*, *Ideation*, *Indicator Behavior* or *Attempt* and bias metrics (*bias mean*, *bias TF-IDF*).

Online language can be used to determine suicide risk. This was framed in CLPsych 2019 Shared Task as a multi-level classification problem with Supportive and Ideation posts being easier to classify and Behavior and Attempt overlap confusing (Zirikly et al., 2019). Past literature concentrates on precision not representational bias which is precisely analyzed in this project

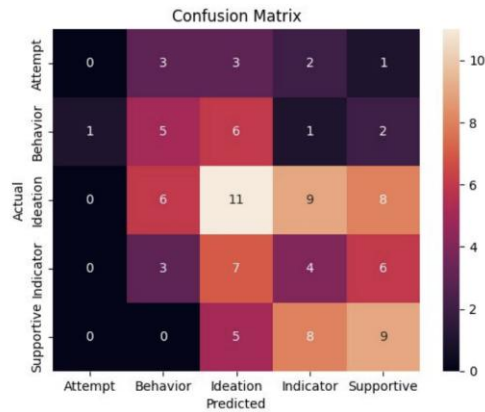
The dataset consists of 500 anonymized posts on Reddit, one post per user, and five categories. Posts are brief, informal and have actual statements of distress. Linguistic cues are conserved with minimal cleaning. TF-IDF vectors are passed through a multiclass Logistic Regression model.

As Figure 1 illustrates, Attempt posts were never correctly classified with zero on the diagonal, while Supportive posts performed best with 9 correct predictions. Ideation and Indicator show the highest confusion, frequently misclassified as each other due to overlapping language. Figure 2 shows that Ideation posts are most common while Attempt posts are least frequent, explaining the model's poor performance on that category. As shown in Figure 3, Supportive posts show the highest bias while Ideation posts show the least, with Behavior, Attempt and Indicator falling in the middle range. TF-IDF weighting partially balances affective signals, reducing but not eliminating bias across all categories.

These results suggest that existing NLP models might overweight ideation language and undervalue real behavioral risk, which can lead to the classification of high-risk posts as false. The controversy between the mean-based and TF-IDF representations represents the conflict between the discriminatory ability and bias reduction. To detect suicide risk accurately, bias-aware feature presentation and consideration of high-risk classes over overall accuracy are necessary to eliminate the critical failures in crisis detection

Appendix

Figure 1 shows Confusion matrix of TF-IDF classifier described in paragraph 4.



Attempt: attempt, somebody, suicide attempt, failed
 Behavior: cutting, hospital, felt, kill im
 Ideation: thank, depression, mental depression, mental
 Indicator: behavior, hyperactive, hyperactive behavior, haha
 Supportive: think, sounds, therapist help, chocolate

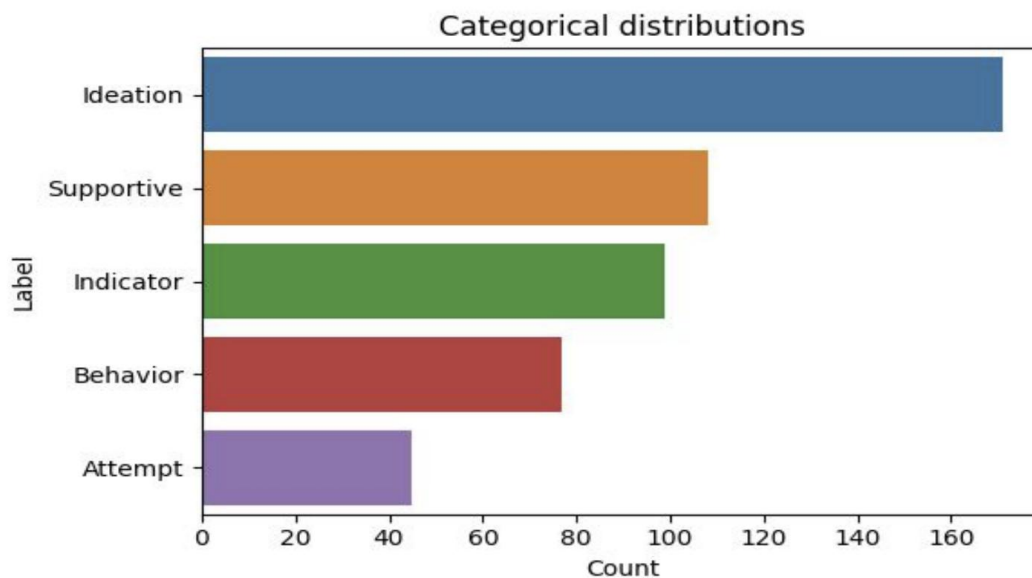


Figure 2 shows Class distribution of suicide-related Reddit posts described in paragraph

4.

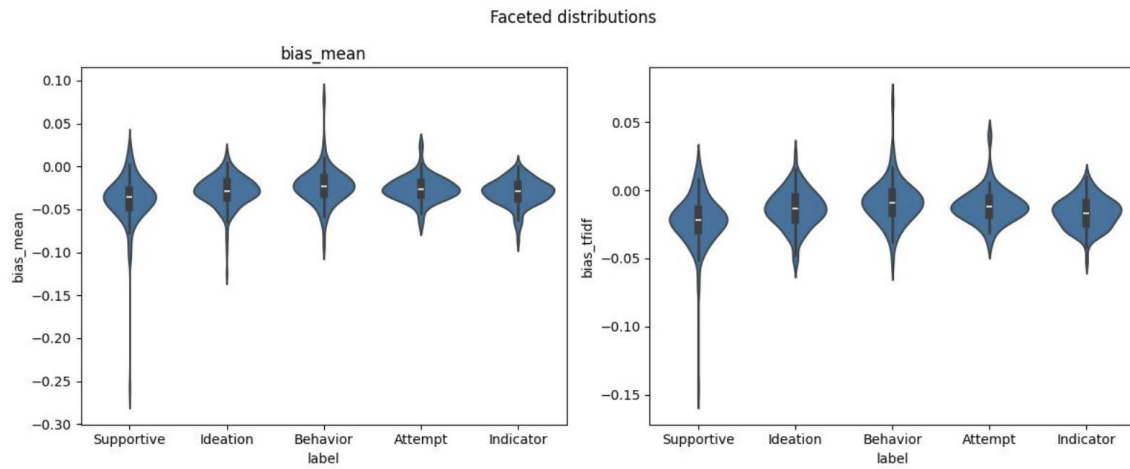


Figure 3 shows a violin chart of TF-IDF Bias Scores in Hopelessness-Support Classification described in paragraph 4.

Reference

Zirikly, A., Resnik, P., Uzuner, Ö., & Hollingshead, K. (2019). *CLPsych 2019 Shared Task: Predicting the degree of suicide risk in Reddit posts*. Proceedings of the Sixth Workshop on Computational Linguistics and Clinical Psychology, 1–12.

Reddit mental-health posts dataset from Kaggle — *Text Summary with Reddit Mental Health Posts* (<https://www.kaggle.com/code/shriyapr/text-summary-with-reddit-mental-health-posts/input>).